



Mapping fire extent and burn severity in Alaskan tussock tundra: An analysis of the spectral response of tundra vegetation to wildland fire

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ABSTRACT

Observed warming in the high northern latitudes has led to an increase in fire occurrence across North American tundra. Our ability to effectively monitor ecosystem change and the carbon cycle in this region depends upon the development of robust and reliable methods of extracting information about fire events and the characterization of fire impacts from remotely-sensed data. Previously developed and routinely applied remote sensing methods yield poor results in assessing the extent and severity of burning in tundra. Here we present an analysis of temporal trends in spectral signatures of burned and unburned areas of tussock tundra obtained from Landsat imagery within six fire events of the 2010 fire season in the Noatak Nature Preserve in Alaska. The analysis revealed a large variability in surface reflectance of burned and unburned areas across space and time. We established that spectral signatures of fire scars can be well separated by both single- and multi-date mapping approaches immediately after burning. However, the spectral signature of burned areas in tundra deteriorates rapidly and, in our study, is poorly distinguishable by the end of the first post-fire season. Our results showed that common mapping methods based on the normalized burn ratio were inferior to other spectral indices (Tasseled Cap Greenness and Brightness) and single Landsat bands (near-infrared) in separating burned and unburned areas as well as in mapping burn severity.

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1. Introduction

High northern latitudes of the North American continent are rapidly changing under the influence of a warming climate. This warming has been accompanied by an earlier onset of spring melt (Markus et al., 2009) and greening of the arctic (Goetz et al., 2005; Myneni et al., 1997), as well as vegetation shifts connected to changing temperatures and snow extent (Bjork & Molau, 2007; Bret-Harte et al., 2001). Climatic conditions are known to be an influence on fire occurrence and fire severity in many regions, including the boreal region of North America where weather-based indices are often used to assess fire severity (Balshi et al., 2009; Flannigan et al., 2005) based on empirically-derived relationships used to develop these weather indices (Van Wagner, 1987). For the fire-prone boreal forest, predictions based on future climate projections show burned area in Alaska's boreal regions will increase by approximately 3.5–5.5 times by 2100, with summer fuel moisture and July temperatures as the main drivers of this increase (Balshi et al., 2009). These same controls are of great influence on burn severity and fuel consumption; so as warming in the high northern latitudes progresses, fires will become larger and more severe (Kasischke & Hoy, 2012). There is increasing evidence of change in fire occurrence and severity in the tundra

areas north of the boreal forests in North America (P.E. Higuera et al., 2011; P. Higuera et al., 2011; Racine & Jandt, 2008). There is also increasing evidence that the historically low-level of burning in most of the tundra region will shift to a more extensive and severe fire regime as temperatures and seasonal moisture conditions are modified across the region (Hu et al., 2010a).

Fire occurrence has significant spatial and temporal variability across Arctic Alaska. Studies have shown that the limits of tundra burning in Alaska are determined by vegetation, topography, and climatic controls, including sea ice retreat and permafrost thawing (Hu et al., 2010a). However, the feedbacks between fire on the landscape, biomass species composition, and regional climate are myriad and complex (Rupp et al., 2001). Fire return intervals range from a few decades up to 5000 years, and tend to be short in regions like the Noatak Watershed and Seward Peninsula, which have higher summer temperatures, lower summer precipitation, and greater fuel availability, in the form of tussock- and birch-shrub tundra (Higuera et al., 2011b). High temperatures lead to reduced fuel moisture, and along with dry, windy weather, enable fire ignition and spread. While only 3% of Alaskan tundra burned between 1950 and 2005 (Higuera et al., 2008), accelerated Arctic climate warming in recent years has led to a marked increase in deciduous shrub density (Sturm et al., 2001); and this fuel change trajectory toward shrub expansion, under a projected 3–5 °C temperature increase could, several centuries from now, result in fire regimes that bear little resemblance to

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those we see today (Higuera et al., 2008). Anecdotal evidence of a changing fire regime is the 2007 Anaktuvuk River fire, which consumed more than 100,000 hectares of tundra on the North Slope (Jones et al., 2009), and was unprecedented in at least 5000 years (Hu et al., 2010a).

The need for satellite based methods of burned area and burn severity mapping for all ecosystems and specifically for remote and inaccessible areas is well recognized in the scientific and management communities (Racine et al., 2006), and this has culminated in development of operationally available severity products for the US that are derived from Landsat using the dNBR and RdNBR methods (Eidenshink et al., 2007). While first evaluations of the success rate for operationally accepted methods showed promising results (Allen & Sorbel, 2008), recent studies have established that these methods do not provide an accurate assessment of burn severity for many ecosystems in high northern latitudes (French et al., 2008). Poor relationships between the field measurements of burn severity and operationally applied algorithms have been reported for boreal wetlands and forests (Hoy et al., 2008; Murphy et al., 2008) and in tundra ecosystems where the single-date version of the dNBR was tested (Boelman et al., 2011). These findings have spurred a number of studies aimed at developing a better understanding of the reasons for the failure of the operational methods and evaluation of the suitability of various spectral indices for developing strong relationships between field measurements of burn severity and satellite-based information. Verbyla et al. (2008) showed that low solar zenith angles in Alaska exacerbate the impact of topography on spectral signatures of burned and unburned areas. These impacts, combined with impacts of phenological change, make reflectance-based indices insufficient in determining the burn severity levels with a required level of confidence. In addition to low solar zenith angles, the short duration of the snow-free season with minimal cloud cover and low concentrations of aerosols over Alaska have been identified as factors limiting the accuracy of coarse resolution burned area mapping approaches (Loboda et al., 2011). Other studies aim at evaluating the performance of additional indices other than the traditionally applied normalized burn ratio (NBR) (Lopez Garcia & Caselles, 1991) and evaluate the impact of spatial resolution on the ability to map burn severity (Boelman et al., 2011). Although important and informative, these efforts have not explored the basics of post-fire changes in spectral reflectance within different spectra of electro-magnetic energy commonly used in optical remote sensing.

A major task of this study is to advance the fundamental understanding of fire-induced changes in remotely sensed surface reflectance. This is accomplished by continuing the analysis of impacts of fire occurrence in Landsat Thematic Mapper (TM) and Enhanced TM Plus (ETM+) imagery. These instruments have been well-developed for monitoring forested ecosystem types (Key & Benson, 2006). The primary goal of this study is to use this Landsat-based assessment to develop robust and ecologically meaningful methods for both mapping the extent of fire impacts and to assess burn severity in tussock tundra. Development of these methods within the tussock tundra types is essential if we are to quantify the potential change in fire affects in this remote and vast region.

2. Study area

The study area is located within the tussock tundra of the Noatak National Preserve in Alaska (Fig. 1), part of the United States National Park System. The preserve is located within the Noatak River basin between the Baird and DeLong Mountains of the Brooks Range above the Arctic Circle. This tundra region of North America has experienced an increase in fire occurrence in the last decade which culminated in 2010, when the largest number of fires burned since record keeping began in the 1950s (Higuera et al., 2011b). Researchers estimate that during the 2010 fire season 37 individual fire events

occurred in the Noatak tundra burning more than 43,180 ha (Higuera et al., 2011a). Paleorecords show that fire is a common ecological disturbance agent in tussock-shrub tundra, with a mean fire return interval of 134 years over the past 2000 years, which occupies warmer and drier areas compared to cooler areas with low-shrub tundra (Hu et al., 2010b).

3. Data and methodology

This study is designed to evaluate the temporal changes in surface reflectance of fire-affected areas in tussock tundra. Here we address three major research questions: 1) what spectral signatures, currently available from the Landsat TM and ETM+ instruments, provide the most optimal differentiation between burned and unburned areas; 2) how well can the results of case studies be extrapolated across tussock tundra; and 3) what spectral signatures from Landsat TM and ETM+ data have the strongest relationship to field assessments of burn severity and how reliable can those assessments be. The methods are therefore designed to develop a better understanding of fire-induced changes to surface reflectance and to produce the basis for future algorithm development in support of fire management in tundra regions of high northern latitudes. The first and second research questions were addressed exclusively through the analysis of a temporal stack of Landsat imagery (described in Sections 3.2–3.4). The analysis of spectral signatures in relation to burn severity was based on available field measurements of burn severity collected by two independent field teams in the 2010 Noatak fires (described in Section 3.1).

3.1. Field data

Field data collections were conducted in the summer of 2011 at five sites (total of 65 plots) that burned in the summer of 2010 within the Noatak Preserve. Data from two field campaigns were used in this analysis, one collection conducted by National Park Service (NPS) personnel lead by fire ecologist Jennifer Barnes, and one led by the authors (French and Loboda). The NPS team visited four of the sites in this study from June 23 to 25, 2011 (Makpik Lake was not sampled); our team collected data exclusively in the Kaluktavik River burn from August 14 to 16, 2011 (see Fig. 1).

The teams collected data for several purposes, including identification of site type, burn severity, vegetation regrowth, fuel consumption, and soil moisture; the relevant data for this study is burn severity. In the field, severity is defined as an assessment of the degree of impact the fire had on the site and on site ecology (Lentile et al., 2006). In general practice, severity is determined using a variety of metrics (French et al., 2008). For this study, severity was determined using ocular (visual) assessment methods developed for tundra types because of restrictions of the field collection permit. Due to significant limitations posed by access to the sites (i.e. availability of landing point within the walking distance of the fire scar) and the difficulty of movement across tussock tundra, the locations of the sites were not spread across the entire burn site. However, effort was made to select sites within a variety of observed burn severity conditions as well as within unburned areas with the aid of air photos and satellite images.

The data collected by the NPS team included severity assessed at four to seven plots per fire site with the minimum distance of 140 m from one site to another using the Composite Burn Index (CBI) as modified for Alaskan Tundra. The CBI is a ground-based plot methodology utilized to field-verify remotely-sensed burn severity measures (Key & Benson, 2006). CBI plots are used throughout the NPS as a tool for validating the satellite-derived estimates of burn severity. CBI ratings can be taken to assess the overall plot burn severity. CBI is collected based on ocular estimates of the degree of

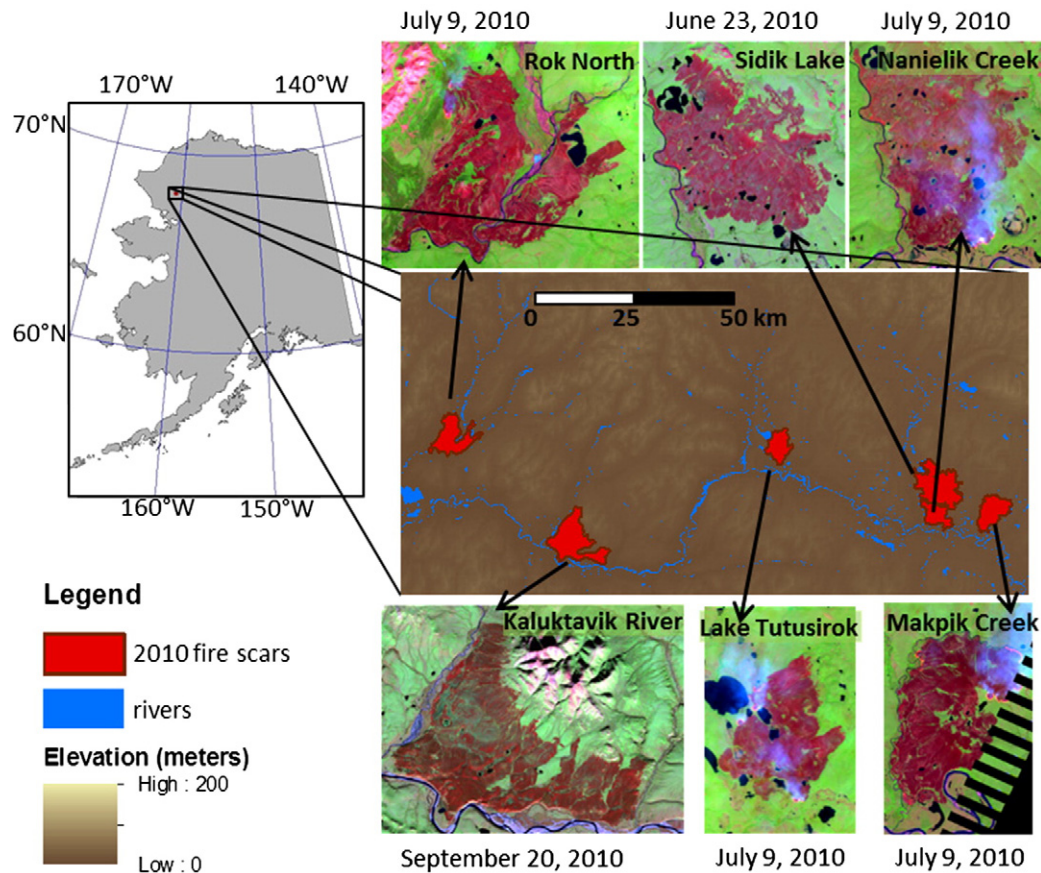


Fig. 1. Location of the study area in the Noatak Nature Preserve in Alaska, USA and the distribution of fire scars within the study area. Insets show individual scars in Landsat imagery (bands 7:4:3 as the R:G:B combination): three different images were used to show the scars, image acquisition dates are shown adjacent to the insets.

environmental change caused by fire to surface features and vegetation layers (strata). Version 4.0 of the CBI form was used.

Assessment of severity at the Kaluktavik site used a modified CBI approach also based on ocular assessment. The fraction of area of four levels of burn severity was determined by the observer within six 10×10 m plots that were randomly located within each of the five sample locations within the Kaluktavik burn. The plots were located at a minimum distance of 50 m away from each other. The severity assessment at Kaluktavik used a four-point scale (unburned, low, moderate, and severe) similar to established methods (Dyrness & Norum, 1983; Viereck et al., 1979). This method was chosen over a rigorous CBI method to save time in the field. In this assessment approach: unburned areas show no sign of being subject to flames with intact herbaceous and shrubs left from before the burn; low severity is indicated by singed moss with post-fire regrowth of vascular plants present, some loss of herbaceous and woody stems but no discernible consumption of the moss or duff; moderate severity is indicated by some ground layer material consumed and all shrubs burned and partially consumed; and severe burning is indicated by consumption of the duff layer and no post-fire re-sprouting of shrubs and inter-tussock vascular plants. Note that the tussock sedges in this type of landscape nearly always survive fire, so will re-sprout even when the inter-tussock material is completely consumed; therefore even in the severe category some re-sprouting from the tops of the tussocks was present. The unburned and low categories showed very little to no surface char while the moderate and high levels included char on the surface. Since all these sites were sampled the year after burning, post-fire greening was extensive due to the ability of sedge tussocks to survive fire.

The Burn Severity Index (BSI) used to compare to remote sensing data was determined from the field data by converting the qualitative

fractional assessment of burn severity within 10×10 m plots, divided into unburned, low, moderate, and severe categories, to a single value following the equation:

$$\text{BSI} = 1 * \text{fraction unburned} + 2 * \text{fraction low} + 3 * \text{fraction moderate} + 4 * \text{fraction severe}$$

The resultant plot of BSI values can range from 1 (100% unburned) to 4 (100% high severity burn) while allowing for a continuous scale of potential burn severity values over a spatially contiguous unit of surface.

3.2. Landsat data pre-processing

Available scenes from Landsat TM and ETM+ with < 30% cloud cover for three summer seasons, pre-fire summer of 2009, fire summer of 2010 and post-fire summer of 2011, covering paths 79 and 80 and row 12 were obtained from the USGS archive (Table 1). The scenes were converted to surface reflectance through the LEDAPS processing system (Masek et al., 2006). Scenes from path 79 row 12 were georegistered to scenes from path 80 row 12 with a maximum root mean square error of 0.014. All scenes were subset to the extent of the six fire scars from the summer of 2010. The scenes were then processed to isolate cloud free vegetated surfaces using a suite of band thresholds chosen visually by the analyst (Fig. 2). These thresholds were designed to be minimally restrictive, i.e. allowing for mapping a more complete extent of potential signal contaminants, to ensure that the subsequent analysis includes spectral signatures coming exclusively from clear surfaces. All resultant masks were visually inspected by the analyst with particular attention to burned areas and areas immediately surrounding the burns to confirm the absence

Table 1

List of Landsat TM and ETM + scenes included in the analysis.

Instrument	Path	Row	Date
TM5	80	12	Jun 4, 2009
TM5	80	12	Jun 20, 2009
TM5	79	12	Aug 16, 2009
ETM	80	12	May 30, 2010
TM5	80	12	Jun 23, 2010
TM5	80	12	Jul 9, 2010
TM5	79	12	Sep 20, 2010
TM5	80	12	May 25, 2011
TM5	79	12	Jun 19, 2011
ETM +	80	12	Sep 22, 2011

of contamination from cloud, shadow, water, flaming front, and bright non-vegetated surfaces.

The spatial distribution of cloud cover varied greatly throughout the 2009–2011 period (Table 2). Only one fire, the Kaluktavik River burn, could be consistently observed in the entire sequence of Landsat scenes. Consequently, we chose to provide a different assessment of the change in spectral signature for each of the six scars based on the available clear surface imagery, and then to compare signatures of individual scars to define a general trend (or lack thereof) in fire-induced changes to surface reflectance in tussock tundra.

3.3. Acquiring burned and unburned surface and burn severity spectral samples

The sample burns in tussock tundra within our study area exhibit a distinct and clearly identifiable signature in the Landsat data (Fig. 1). Separating burned from unburned area based on a single-band or two-band spectral index threshold may bias the subsequent

analysis of spectral signature variability towards the selected bands. To avoid biasing our analysis we used the Spectral Angle Mapper (SAM) (Kruse et al., 1993) – a supervised classification approach which ingests spectral information from all Landsat bands to divide spectral space into burned and unburned samples. This method compares the unclassified spectra in the imagery to a known sample (i.e. training data) by measuring the angle between the vectors of the spectra in multidimensional space. We limited the extent of mapping for each scar to a rectangular box surrounding the burn in the image, the box was defined by the analyst, who set the spectral angle for all bands not to exceed 0.15 radians. All classifications were subsequently visually inspected by the analyst, and areas where the analyst could not confirm the results of classification were manually eliminated.

The goal of subsequent pre-processing steps was to extract the clear samples of burned and unburned areas within and around the scars. Sample scars in our study area burned at different times and therefore presented the best identifiable signature in different Landsat scenes (Fig. 1). Table 2 indicates the dates of burning determined from the MODIS active fire product (Giglio et al., 2003) and mapping for each individual scar in the analysis. Visual analysis has shown that all scars were most identifiable in Landsat scenes from June 23, 2010 for the Sidik Lake burn (burning was completed before June 23, 2010) and July 9, 2010 for Rok North, Lake Tutusirok, Nanielik Creek, and Makpik Creek burns. Most of the Kaluktavik River burn was mapped from the September 20, 2010 scene because most of the area burned after July 9, 2010. However, the portion of the Kaluktavik River burn which burned before July 9, 2010 (as visible from the July 9, 2010 image) was not well mapped by the SAM algorithm and thus was excluded from further analysis. The Rok North, Lake Tutusirok, Nanielik Creek and Makpik Creek fires started burning prior to July 9, 2010 but still had on-going burning on that date. We eliminated all areas of on-going burning and areas which burned after the July 9, 2010 (using the image from September 20, 2010 as a reference) from the burned/unburned classification and included only areas which had completed burning by July 9, 2010. Similarly, all unburned areas around the scars were limited to areas which were known to be unburned throughout the full time-span of the study from June 4, 2009 to September 22, 2011 and were surrounded or immediately adjacent to the burns. In total we identified the following number of burned and unburned pixels for each scar: 1) Rok North – 9652 burned, 10,013 unburned; 2) Kaluktavik River – 8262 burned, 1467 unburned; 3) Lake Tutusirok – 5857 burned, 9512 unburned; 4) Nanielik Creek 4136 burned, 6896 unburned; 5) Makpik Creek – 18,256 burned, 4798 unburned; 6) Sidik Lake – 17,795 burned, 13,312 unburned. This approach allowed us to obtain samples of burned and unburned areas for each of the six scars using a consistent, replicable, and relatively unbiased approach with verification from the analyst.

Clear surface masks for each individual date of Landsat imagery were applied to the burned/unburned samples and the multi-date clear-view sequences of imagery were identified for each scar. Only one scar (the Kaluktavik River burn) has a sufficient sample of burned and unburned area visible in all Landsat images (Table 2). A random sample of 1000 burned and 1000 unburned pixels for each scar was obtained from the multi-date clear-surface view to extract samples of surface reflectance in individual Landsat bands, including blue (band 1), green (band 2), red (band 3), near-infrared (NIR – band 4), shortwave infrared around 1.6 μm (SWIR 1.6 – band 5), and SWIR around 2.2 μm (SWIR 2.2 – band 7), to form burned and unburned surface reflectance samples and, subsequently, samples for a suite of spectral indices.

We selected a suite of spectral indices which are commonly used in characterizing surface parameters with relevance to burned area mapping. While this ensemble of spectral indices is not exhaustive, it is representative of various approaches to burned area mapping and combinatory capabilities of various Landsat bands. Our selected suite of spectral indices includes the normalize difference vegetation

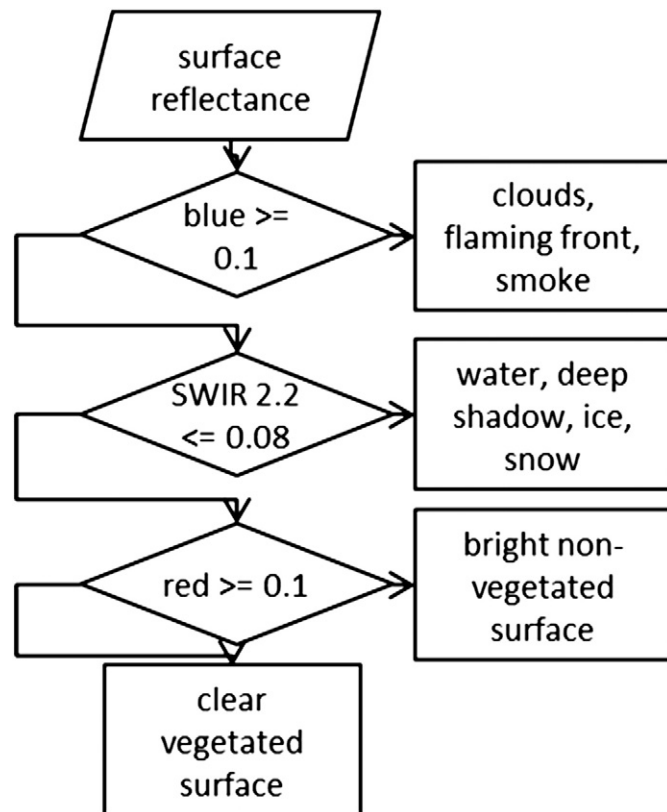


Fig. 2. Data pre-processing flow to separate clear vegetation surfaces for spectral signature analyses. B1, b3, and b7 represent Landsat bands 1 (blue), 3 (red), and 7 (SWIR 2.2), respectively.

Table 2

Clear surface views and timing of burning for individual scars within the analysis. Plus “+” denotes the image date where clear surface observations were available, “X” indicates the image date for determining burned and unburned sample areas.

Scars	Burn dates, 2010	Jun 4 2009	Jun 20 2009	Aug 16 2009	May 30 2010	Jun 23 2010	Jul 9 2010	Sep 20 2010	May 25 2011	Jun 19 2011	Sep 22 2011
Rok North	Jul 1 - 9	+	+	+	+	+	X	+		+	+
Kaluktavik River	Jul 2 - 16	+	+	+	+	+	+	X	+	+	+
Lake Tutusirok	Jul 2 - 10	+	+	+	+	+	X	+		+	
Nanielik Creek	Jul 4 - 10	+		+		+	X	+		+	
Makpik Creeks	Jul 3 - 10	+		+		+	X	+		+	
Sidik Lake	Jun 9 - 11	+		+		X	+	+	+	+	

index (NDVI) (Tucker, 1979), normalized difference SWIR index (NDSWIR) (Gerard et al., 2003), NBR (Lopez Garcia & Caselles, 1991), and brightness, greenness, and wetness components from the Tasseled Cap index (TCB, TCG, and TCW, respectively) (Crist & Cicone, 1984). Each of the indices presents some form of explanatory power in which two or more bands reveal changes in vegetation or other surface parameters, as can be expected after burning. The first three indices assess the normalized differences between reflectance in NIR as the first component and red, SWIR 1.6, and SWIR 2.2 for NDVI, NDSWIR, and NBR, respectively, as the second component. These indices use the following general equation:

$$\text{Index} = (\text{NIR} - \text{component 2}) / (\text{NIR} + \text{component 2})$$

Tasseled cap indices are calculated by transforming the spectral contribution from all surface reflectance bands (1–5, 7) through a series of weighting coefficients, specified for Landsat bands (Crist, 1985). For burn severity assessment, in addition to all indices listed above we also included relative dNBR (RdNBR) which has been shown to be more closely linked to burn severity in low biomass environments (Miller & Thode, 2007). RdNBR is calculated as:

$$\text{RdNBR} = \text{dnbr} / \text{SQRT}(\text{ABS}(\text{prefireNBR}/1000))$$

Finally, we extracted surface reflectance samples over the field points where burn severity data was collected during the field data campaigns described earlier. These surface reflectance signatures and the derived spectral indices were subsequently used in the assessment of burn severity mapping capabilities from remotely-sensed data.

3.4. Spectral signature analysis methods

Several sets of statistical techniques were applied to analyze the separability of burned and unburned spectral signatures and burn severity measurements. The first analysis was designed to assess the uniformity of unburned vegetation signatures across the entire span of the Landsat scene. Within this analysis we compared unburned samples of surface reflectance and spectral indices of the six scars within three scenes with the lowest amount of atmospheric contamination (from June 23, 2010, September 20, 2010 and June 19, 2011). The combination of one-way Analysis of Variance (ANOVA) and the post-hoc Scheffe tests were used to examine the uniformity of spectral signatures among the six sample areas within the Landsat scene and identify the specific areas with the greatest differences in unburned vegetation.

We then compared the burned and unburned samples for each individual scar at different dates using a Student's t-test to determine if samples were statistically different and measured the magnitude of this difference in terms of T-statistic magnitude. Additionally, we evaluated the separability of burned and unburned samples in individual bands and spectral indices using the mean (μ) and standard deviation (σ) differencing. In this approach we evaluated how far apart the means of the two populations were expressed in standard

deviations. Specifically, we checked if the mean of burned (μ_b) was 1, 1.5, 2, or 3 standard deviations apart from the mean of the unburned sample (μ_u) following the expression: if $\mu_b > \mu_u$ then assess if $(\mu_b - a * \sigma_b) > (\mu_u + a * \sigma_u)$ otherwise assess if $(\mu_b + a * \sigma_b) > (\mu_u - a * \sigma_u)$, where a is expressed as 1, 1.5, 2, and 3, σ_b and σ_u are standard deviations of burned and unburned samples, respectively.

The final test of binary “burned/unburned” signature separability included a comparison of pre-burn to post-burn changes (i.e. delta) of surface reflectance in individual bands and spectral indices for burned and unburned samples. In this test we assessed the difference of signatures in all combinations of pre-burn and post-burn mapping dates of individual scars using the Student's t-test statistic and the mean and standard deviation differencing approach described above.

The relationship between burn severity and surface reflectance was assessed using linear regressions between the field assessments of burn severity converted to a single severity index – BSI (described in Section 3.1.) and surface reflectance in individual bands or spectral indices acquired from single-date images as well as differenced values from various combinations of pre- and post-burn images for each individual scar. For single-date assessments, we assessed regressions for pre-burn and post-burn images to evaluate potential for impact of initial vegetation distribution and condition on post-burn severity. In addition, because of the difference in the time of burning for individual scars, we excluded BSI measurements from the Sidik Lake burn (which burned in early June) from the regression for the June 23, 2010 image, which was subsequently considered a pre-burn scene. All scars, with the exception of the Kaluktavik River burn, were undergoing burning or have completed burning in most sections by July 9, 2010. Subsequently, we removed the Kaluktavik River burn BSI measurements from the regression for July 9, 2010, which is considered the first post-burn date in the regression analysis.

For delta (pre-burn – post-burn) conditions we computed regressions for all possible combinations (25 total) of pre-burn (June 9, 2009, June 20, 2009, August 16, 2009, May 30, 2010 and June 23, 2010) and post-burn (July 9, 2010, September 20, 2010, May 25, 2011, June 19, 2011 and September 22, 2011) omitting points for the Sidik Lake burn from all assessment for the pre-burn date of June 23, 2010 and all points for the Kaluktavik River burn from all assessments from the post-burn date of July 9, 2010.

4. Results

4.1. Assessment of spectral similarity of unburned tundra under clear-sky conditions

We explored the variability in surface reflectance across the sections of tussock tundra using the clear-sky unburned samples acquired from Landsat data. Our main goal was to evaluate if the pre-fire (i.e. unburned) conditions differed across the burned areas included in the sample and thus could have contributed to the differences in burning processes across the area. The results of the clear-sky comparison of unburned samples clearly showed that the unburned samples across the study area were statistically significantly different for samples near various scars (Appendix A). The

ANOVA F-statistic showed a large difference between samples of unburned area for all clear-sky images with the f values far exceeding the critical F-value of 4.109001 at $p < 0.01$. Overall the blue, green, and red bands and TCB index show the largest magnitude of dissimilarity in unburned samples compared to other bands and indices. The comparatively large dissimilarity in the visible bands could be explained at least in part by the general sensitivity of visible bands to variation in atmospheric effects across the image. Although the LEDAPS procedure removes much of the atmospheric influence on surface reflectance, those effects are not completely removed but rather significantly diminished and thus still detectable within the shortest wavelength bands – Blue, Green, and Red. This is particularly likely considering that many of the scenes we acquired had a considerable amount of cloud cover and thus, although we removed direct impacts of visible clouds on surface reflectance in the preprocessing stages, we would not be as successful in removing the impact of thin cirrus clouds and small-scale variability within atmospheric column over various sections of the image. The magnitude of these differences and the sensitivity of specific indices vary based on the timing of image acquisition. There is also considerable variability in ANOVA F-value magnitude of surface reflectance samples within the NIR, SWIR 1.6, and SWIR 2.2 bands and subsequently in the spectral indices which normalize the spectral information from these bands – NDVI, NDSWIR, and NBR. The TCG index appears to have the most consistent ANOVA F-value for unburned area comparison at different times.

F-values of post hoc Scheffe tests attribute the ANOVA-determined dissimilarities of several samples of unburned area to differences between specific samples. The results, summarized in Appendix A, show that unburned areas within samples collected within and around Lake Tutusirok, Nanielik Creek, and Makpik Creek burns (all located in the East side of the Landsat images) are more similar to each other than to any other scars. It is surprising, however, that unburned areas in and around the Sidik Lake burn, (flanked by the Nanielik Creek burn in the south) appear to be no more similar to those adjacent to the Nanielik Creek burn than it is to those near the Rok North burn (the left-most scar within the Landsat image). The differences among unburned areas were most pronounced around the Kaluktavik River and Sidik Lake burns in all scenes with a greatest magnitude of differences in the scene from June 19, 2011. Similarly to ANOVA results, the largest Scheffe test F-values, showing the greatest dissimilarity between samples, were found within the blue, green, and red bands. F-values for all spectral indices were considerably lower; however, no single index appeared to provide a consistent (i.e. across all scars) and stable (i.e. across the three clear-sky images) assessment of differences between unburned samples within our study area.

The findings of the ANOVA and Scheffe tests indicate that there is a large variability in local surface conditions across a relatively small (less than 180 km) span of similar treeless tundra which preclude the broad application of a local case (e.g. individual scar-based) analyses to the region with similar land cover types. A large sample spread across the extent of the specific land cover within the region is required to develop an approach to characterizing the surface conditions within tundra. No single index provides a superior ability to differentiate or unify the samples from various locations across space/time. Additionally, in some cases even spatial adjacency of two samples does not guarantee their spectral similarity.

4.2. Spatial and temporal patterns in burned and unburned spectral signature separability

This analysis contains two parts. The first is focused on analyzing the single-date separability of burned and unburned samples for individual scars and generalizing the findings for all samples. The second part focuses on the analysis of differenced (or delta) pre-burn minus post-burn spectral change assessment.

Student t-test analysis of the burned and unburned samples from each of the scars was carried out for different dates (Appendix B). The results show that even though there is a statistically significant difference between the samples of burned and unburned areas within and around the same scar prior to the occurrence of the fire, the differences are very small compared to those sampled after the fire occurrence. None of the pre-burn samples were more than the mean plus or minus one standard deviation apart from each other. The greatest difference in the majority of scars is observed immediately after the fire occurrence within TCG, NIR, and NBR while TCW and SWIR 2.2 offer the lowest separability of burned and unburned samples. It is particularly important to note that, as has been shown in previous studies (Boelman et al., 2011), the response of the spectral signature in SWIR 2.2 is highly variable across both space and time. In some cases burned pixels in the SWIR 2.2 band have a higher reflectance than in unburned areas (thus resulting in positive T-values) while in other cases the reflectance in SWIR 2.2 is lower than unburned areas (thus resulting in negative T-values). Additionally, reflectance of SWIR 2.2 changes in opposite directions over time, indicating that SWIR 2.2-dependent indices (e.g. NBR) will also vary considerably in their mapping accuracy driven by differences in the date of image acquisition relative to the date of the burn. While we did not explore specific reasons for variations in SWIR 2.2 behavior pre- and post-fire across the study area and through time, it is likely that this variability can be partially explained by temporal and spatial variations in moisture content of remaining vegetation and soils linked to local precipitation patterns as well as deposition of organic char (Asner & Lobell, 2000; Boelman et al., 2011). This finding supports previous assessments of NBR spectral stability conducted by Roy et al. (2006) and makes a specific case for limiting NBR use in tundra fire mapping to within the same fire season and before vegetation senescence.

The deterioration of spectral separability between burned and unburned areas in tundra occurs rapidly. Our results show that by the fall of 2010 the magnitude of the T-values drops by a factor of 0.5 to 0.3 from the immediate post-burn assessment. The Kaluktavik River burn represents the exception to these observations partially because it burned later in the season, and also because the first observation after the fire occurrence was done during the senescence period in late September. In this case, the burned/unburned spectral signature separability was reduced only slightly after the first post-fire date and remained at nearly the same level until the fall of 2011 when the T-values dropped by a factor of 0.5 for different indices and spectral bands. For all scars except for the Kaluktavik River burn the first mapping date occurred during the greenest times of the season and the burned areas were highly separable (greater than 3σ from each of the means) in NIR, TCG, and NBR. These were closely followed by NDVI, NDSWIR, and TCB. The separability of signatures in bands 1–3, 7 and TCW index were less than 1σ from each of the means even immediately after the burning occurred. By September 20, 2010 the separability dropped to 1 and 1.5σ from each of the means for all scars in TCB, NIR and SWIR 1.6. Most of the scars were still at least 1σ from each of the means apart in TCG and bands 1–3. However, for NDVI, NBR, and NDSWIR the spectral signatures of most scars became less than 1σ apart. The separability of burned and unburned signatures reduced greatly over the subsequent season (2011) where only TCG and NIR offered more than 1σ distance from each mean for all scars burned in 2010, although five scars were still separable in TCB and four separable in NDSWIR and NBR with greater than 1σ distance. The TCW is the only index which offered no separability of the signatures greater than 1σ for any of the scars in 2011. By the last image used in our analysis sequence TCB and SWIR 1.6 offered 1σ separability for one of the scars and no index or spectral band offered 1σ separability for the other scar visible at that date.

Based on the results of this analysis we can conclude that the selection of spectral indices for burned area mapping immediately post-fire is of lower importance compared to the selection at a later

mapping date if no early post-fire image is available. It appears that fresh burn signatures are easily separated in many spectral spaces with the highest degree of difference in TCG, NIR, and NBR. However, burn separability in the NBR space appears to deteriorate faster than that within TCG and NIR, limiting the success rate of NBR-based methods for burned area mapping in tundra. TCG and NIR offer the greatest difference between the means of burned and unburned samples during all post-burn assessment dates and thus are likely to provide a more stable method of burned area mapping through time from single-date mapping methods.

Many burned area mapping approaches rely on the magnitude of change between pre-burn and post-burn surface reflectance of fire impacted areas. Therefore, our second burned/unburned signature separability assessment focused on differences in selection of pre- and post-burn dates for mapping burned areas. We conducted t-test separability of pre-burn minus post-burn reflectance in all spectral bands and indices for different pre- and post-burn dates for each individual scar. The results of this analysis are summarized in Figs. 3 and 4.

Pre- and post-burn dates in the analysis varied for different scars based on the availability of clear surface view within an individual scene and the timing of burning. For all pre-burn image dates, burned/unburned signature separability peaked in the earliest post-burn image with TCG, NIR, and NBR offering at least 3σ separation between the means of burned and unburned samples for all scars. Signal separability in subsequent images declined by roughly a factor of 0.5 with the strongest mean separability across all scars provided within TCB and NIR spectral spaces. It is difficult to draw strong conclusions because of a very small sample (one to two scars) for May 25, 2011 and because of the considerable variability of signature separability in various spectral bands and indices of individual scars within the study area, however, the NIR band appears to provide consistently strong separation of burned and unburned samples. A more robust sample across multiple scars for the post-burn date of June 19, 2011 suggests that TCG and NIR offer superior differentiation of burned and unburned signatures across multiple scars. The TCG and NIR are closely followed by NBR, indicating that during the full green-on conditions NBR can also provide a strong differentiation between burned and unburned areas. The September 22, 2011 sample is also small and, therefore, does not provide an opportunity to draw conclusive results. However, it appears that in the majority of cases by late September 2011 the signal separability between burned and unburned samples is very low likely due to both considerable vegetation recovery of the burned areas as well as senescence of vegetation across all samples.

Fig. 3 shows the trajectory of mean values in individual bands and spectral indices across time for burned and unburned samples. The figure shows that NIR and NBR have the greatest magnitude of change after the fire occurrence. This magnitude of dynamic range of the NBR signal has been cited by previous studies, including our own previous work, as an indicator of NBR's superior ability to differentiate burned and unburned signatures (Loboda et al., 2007). However, as previous analyses within this paper show, burned and unburned signatures can be separated with a higher degree of confidence by TCG which has a much smaller drop in values from burned to burned samples. This is explained by the low variance in TCG values around the mean ($\sigma = 0.02$ for July 9, 2010) compared to that of NBR values ($\sigma = 0.12$ for July 9, 2010): even though the amplitude of change is greater in the NBR space, the spread of values around the mean implies a greater overlap of burned and unburned pixels in spectral space than that of TCG, where values of individual pixels are tightly clustered around the mean.

4.3. Burn severity assessment

Field assessments of burn severity were collected by several field crews during the summer of 2011. Although the crews followed the

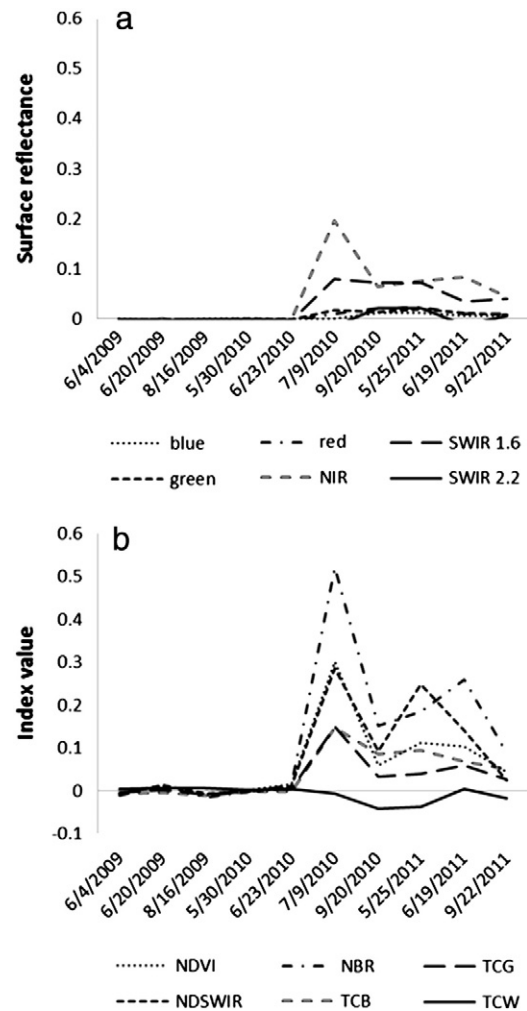
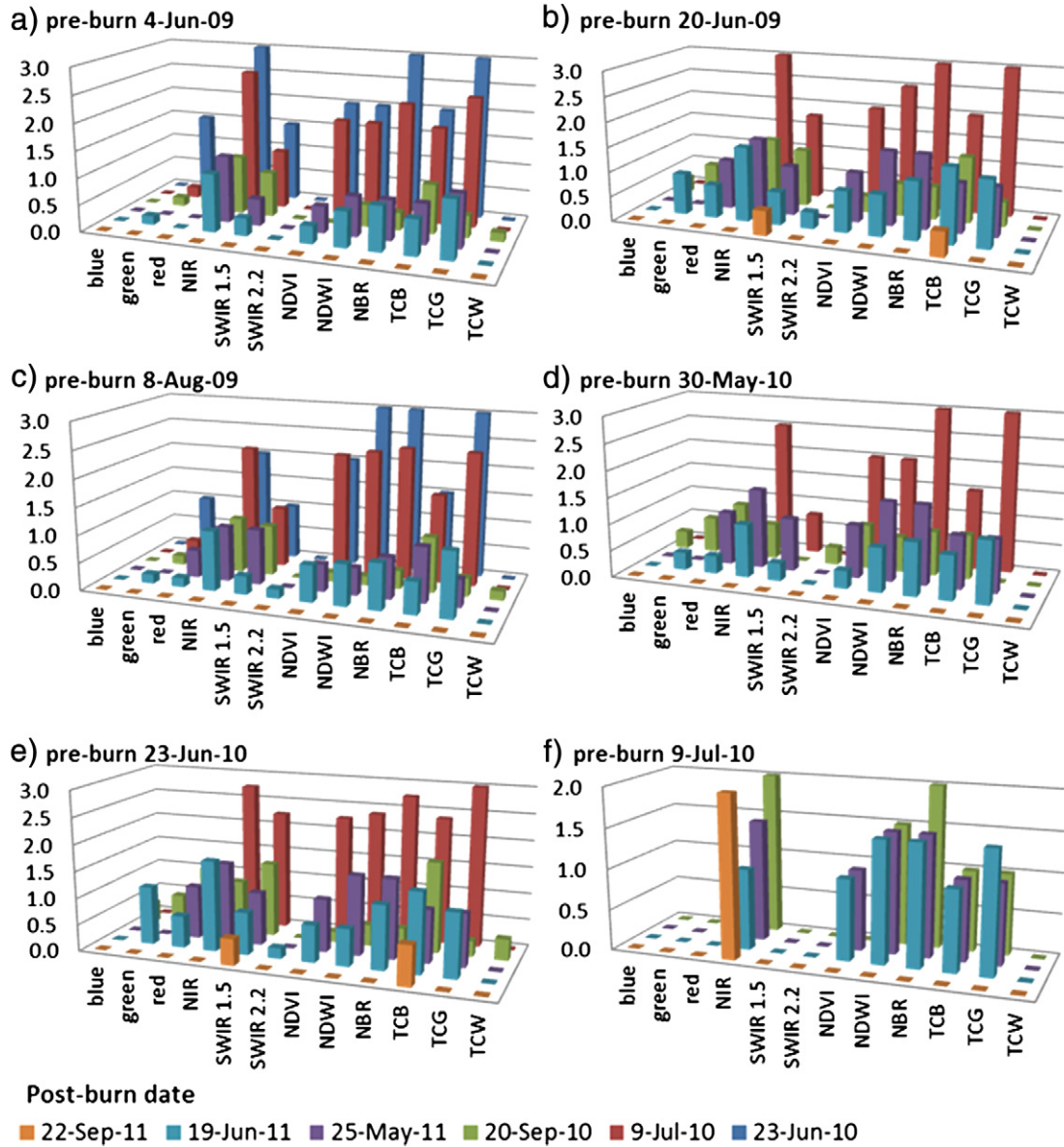


Fig. 3. Mean signal amplitude (unburned – burned) for spectral bands (a) and indices (b).

same data collection protocol, there are large differences in the amount of data collected within various burned areas. Of the total 65 available BSI measurements, 38 were located within and around the Kalukta River burn, while only five, nine, seven, and six points were located in and around the RokNorth, Lake Tutusirok, Nanielik Creek, and Sidik Lake burns, respectively. Subsequently, the analysis is dominated by conditions observed in the Kalukta River burn.

The results of the regression assessment of field observations of burn severity expressed through a BSI index are shown in Fig. 5 and Appendix C. The analysis of these results presents a suite of interesting and important findings. First, the r^2 measures for pre-burn dates show an overall weak relationship between spectral information within individual bands or indices and the burn severity estimates. Although weak, some of the relationships are statistically significant even at the $p < 0.001$ level. Of particular interest are r^2 of 0.24 and 0.23 ($p < 0.001$) in NDVI and red spectrum, respectively, observed during June 23, 2010 or immediately before fire occurrence. The slope of the regressions is positive for NDVI and negative for red implying that burn severity may be linked to the amount of accumulated biomass on the surface. However, further studies would be necessary to evaluate this premise.

Single-date regressions also show that after the burn, the relationships between spectral signatures and BSI are much stronger (up to 0.84 in TCB on May 25, 2011) with $p < 0.001$ for most cases. These relationships are the weakest immediately after the fire occurrence (July 9, 2010) and by the end of the first post-fire season (September 22, 2011). Immediately after the fire (July 9, 2010) the strongest



n(scars)		Post-burn date					
		23-Jun-10	9-Jul-10	20-Sep-10	25-May-11	19-Jun-11	22-Sep-11
Pre-burn date	4-Jun-09	1	5	6	2	6	2
	20-Jun-09	0	2	3	1	3	2
	16-Aug-09	1	5	6	2	6	2
	30-May-10	0	2	3	1	3	2
	23-Jun-10	0	4	5	1	5	2
	9-Jul-10	0	1	1	1	1	1

Fig. 4. Separability of means for burned and unburned samples assessed using delta pre-burn – post-burn metric (see Section 3.4 for details) and expressed in standard deviations (y axis) from respective means. Panels a – f show differences in spectral signature separability for individual pre-burn dates where post-burn dates for individual bands and indices along the x-axis are shown in color. For example, panel “a” shows that for pre-burn date of June 4, 2009 NIR, NBR, and TCG indices offer the best differentiation between burned and unburned samples ($\geq 3\sigma$) when paired with post-burn date of June 23, 2010; and none of the indices show reliable separability for post-burn image from September 22, 2011 ($< 0.5\sigma$). Number of scars in the sample noted in the table below the figure.

relationships are observed between BSI and the measures of greenness – NDVI and TCG, both at r^2 of 0.72 ($p < 0.001$), followed by NIR signature ($r^2 = 0.69$, $p < 0.01$). However, the strongest relationship ($r^2 = 0.84$, $p < 0.001$) was recorded on May 25, 2011 before the full

onset of greenness in the TCB values, closely followed directly by NIR spectrum ($r^2 = 0.82$, $p < 0.001$). TCB, SWIR 1.5, NIR, and NBR held the strongest relationships with BSI during June 19, 2011 (r^2 of 0.81, 0.80, 0.79, and 0.79, respectively, at $p < 0.001$). By the

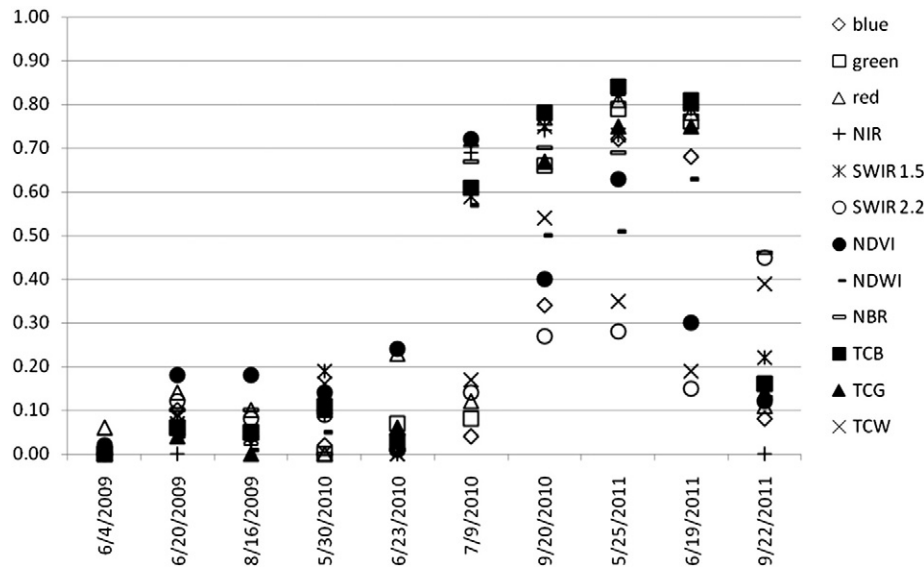


Fig. 5. Strength of relationship (expressed through r^2 of a regression) between burn severity (expressed as BSI) and spectral bands and individual indices. Dates prior to 7/9/2010 show the relationship prior to fire occurrence and date beginning with 7/9/2010 indicate the relationships after the fire occurrence.

senescence of the first post-fire year (September 22, 2011) the relationships again became weak, with NBR at r^2 of 0.46 ($p < 0.001$) reporting the closest regression.

The analysis of relationships between BSI and the delta of spectral signatures in pre- and post-burn images shows no considerable improvement on the predictive capabilities of single-date spectral signatures (Fig. 6 and Appendix D). The closest relationship ($r^2 = 0.87$, $p < 0.001$) was again observed within TCB signature of delta between the May 30, 2010 and May 25, 2011 dates. Similarly to single-date assessments of burn severity, differenced spectral signatures do not perform well at the end of the first-post-fire season (September 22, 2011). During September 20, 2010, May 25, 2011, and June 19, 2011, at least several indices or bands produced reasonably correlated ($r^2 > 0.7$) and significant (at least $p < 0.01$) relationships between

delta signature and BSI. Large differences in r^2 of individual indices were observed for various combinations of pre-burn and post-burn images. No single index produced consistently superior assessments of burn severity across all combinations evaluated in this study: TCB/BSI relationship exceeded r^2 of 0.8 in 6 pre- and post-burn image combinations out of 25, followed by NIR (4 out of 25), SWIR 1.5 and RdNBR (3 out of 25), and NDVI and NBR (1 out of 25). As expected, RdNBR outperformed delta NBR in almost all cases, and was one of the best performing indices overall, although the r^2 in most instances was below 0.8. Moreover, the assessments yielded by individual spectral bands or indices were highly unstable with ranges between minimum and maximum r^2 of 0.6 (SWIR 2.2) to 0.83 (NIR and SWIR1.6). RdNBR offered the greatest mean r^2 (0.67) although it had the strongest relationship ($r^2 = 0.84$, $p < 0.001$) for

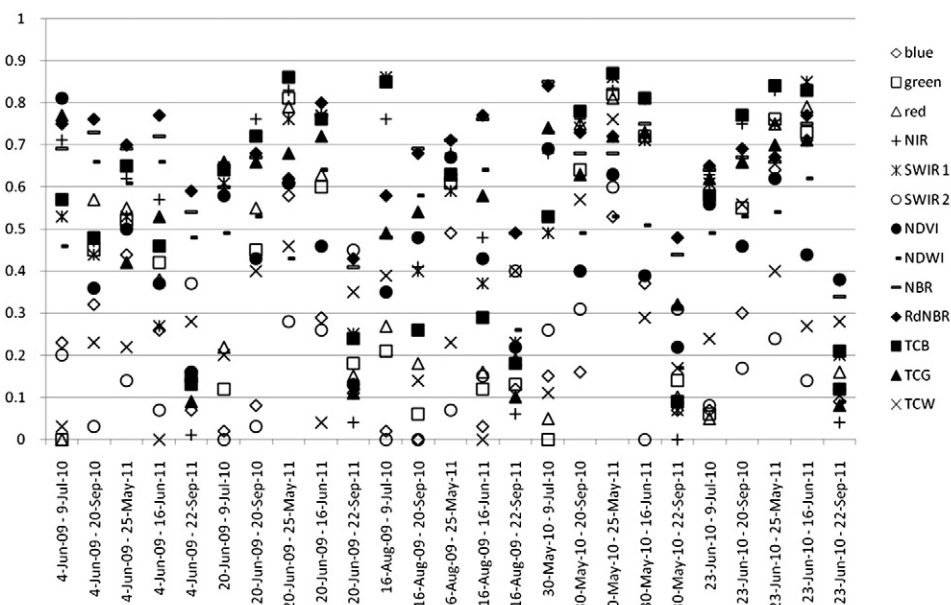


Fig. 6. Strength of relationship (expressed through r^2 of a regression) between burn severity (expressed as BSI) and differenced (pre-burn – post-burn) individual indices and spectral bands.

delta May 30, 2010 – July 9, 2010. It is notable that NBR showed the closest relationship with BSI measurement on the same day ($r^2 = 0.85$, $p < 0.001$) as RdNBR, which was the only NBR r^2 value above 0.8 within this study. On average, the mean r^2 of all regressions were the highest for delta May 30, 2010 – May 25, 2011, June 23, 2010 – May 25, 2011, June 20, 2009 – May 25, 2011, and June 23, 2010 – June 19, 2011 (mean $r^2 = 0.72$, 0.65, 0.64 and 0.64, respectively) when in each case TCB recorded the highest $r^2 > 0.8$.

5. Discussion

The results of this study highlight several important components. First, the timing of image acquisition for single image and multiple-image approaches strongly influences the outcome and success rate for both mapping burned area and burn severity assessments. It is likely that the importance of date of image acquisition is driven by both ecological specifics of tussock tundra, fire ecology, as well as sun-instrument geometry and atmospheric composition. The initial visual analysis of the imagery showed that areas where burning was completed by July 10th have experienced a considerable amount of regrowth by September 20 of the same year when the first clear sky image of the entire area became available. This rapid green-up of vegetation post-fire makes it imperative that early-season scars should be mapped within a month after the fire occurrence as the spectral signature deteriorates quickly. Although we were unable to ascertain the exact speed of signature deterioration within this study based on greater than a month-long gap between subsequent image availability, we have established that by September 20, 2010 many portions of fire scars which completed burning by July 10, 2010 were not spectrally separable from unburned areas.

In addition to the quick recovery of early-season scars, the decrease in signal separability by late September is likely to be driven by the general senescence of tundra vegetation at the end of the growing season. The reduction in contrast between burned and unburned vegetation during senescence is also responsible for masking the effects of burning on the surface in the image from September 22, 2011 where no index or spectral band was able to provide a good separation between burned and unburned sections of tussock tundra. Finally, the low solar zenith angles in late September (compared to those in late May and June) are also likely to contribute to the diminished contrast by amplifying the impacts of topographic and vegetation shadowing as well as lengthening the path through the atmosphere with subsequent increase in potential for atmospheric attenuation of the radiometric signature.

However, even with low solar elevation angles and senescence of vegetation, the burned area within the Kaluktavik River burn, which completed burning in late July 2010, were still reliably discernible in the late September 2010 image. In this case the brighter and more persistent signature may have resulted from a combination of more severe burning and a shorter recovery period before the onset of senescence.

Our results indicate that TCG, NIR, and NBR provide the most reliable separation between burned and unburned areas with the highest degree of separation in the earliest non-senescent scene. While burned/unburned separability diminishes rapidly with time, the NIR and TCG (and to a slightly lower level NBR) do offer at least 1.5σ separability (from each mean) between means of burned and unburned samples through June of the first post-burn year (2011 in this study). By the end of the first post-burn year, no index or spectral band can offer a reliable separation between the burned and unburned signatures. Differencing pre-burn and post-burn images offers marginal improvements in signal separability. Combined with difficulties in finding two clear-sky scenes due to considerable amount of cloud cover, short snow-free period of observation, low solar elevation angles in the beginning and end of the snow-free periods, and the necessary pre-processing of the scenes, delta pre-burn/post-burn mapping

approaches may be more cumbersome than helpful in mapping the extent of burns in tussock tundra.

While greenness-based measurements (TCG and NIR) provide the strongest burn/unburned signal separation, TCB appears to be the best index for burn severity assessment. TCB showed the strongest and closest relationship with burn severity measurements from single- and multi-image assessments ($r^2 > 0.8$). The field plots for burn severity assessment were 10×10 m, and thus cover less than a full Landsat pixel (30×30 m). Although we made a specific attempt to set up transects and the plots within areas of visibly homogeneous burn impact, it is possible that the smaller than pixel field plots were not fully representative of the conditions across the entire pixel and lowered the r^2 values for relationships between spectral bands and indices and BSI. Similarly to the results for burned/unburned signature separability, selection of mapping date has a pronounced effect on the accuracy of burn severity assessment from remotely sensed data. Unlike burned/unburned signatures, burn severity shows closer relationships between BSI and satellite-based indices in late May 2011 rather than immediately after burning when the char signal overwhelms the degree of surface impact. Immediately following the fire event, the highly flammable inter tussock sections composed of senescent herbaceous and woody stems are readily consumed resulting in extensive char deposition on the surface. By the following spring, snowmelt and other precipitation partially remove the char and expose the more reflective remaining duff layer or mineral soil which is subsequently picked up by the TCB index. It is likely that TCB is sensitive to the completeness of combustion of inter-tussock material which increases the overall brightness of the surface no longer covered by the senescent vegetation.

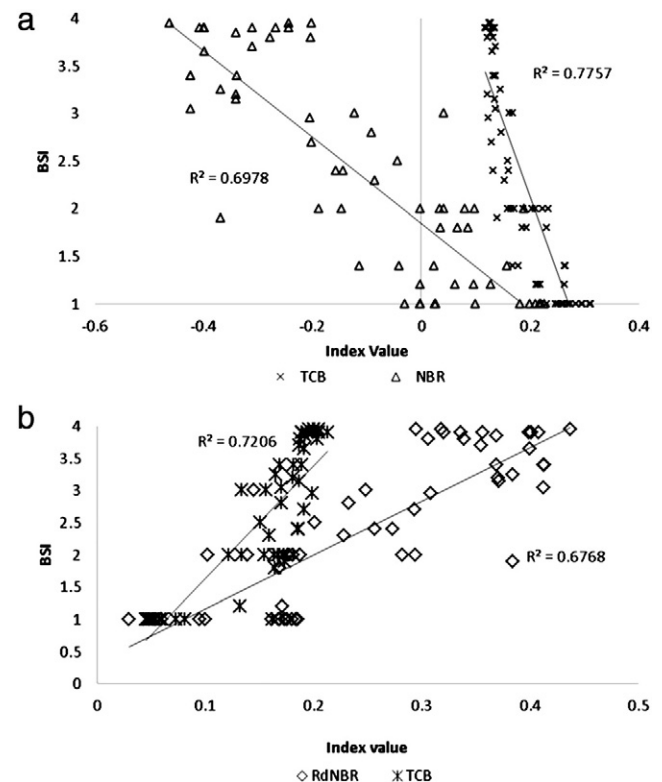


Fig. 7. Distribution of individual points for spectral indices as a function of BSI for: a) single date TCB and NBR for September 20, 2010 ($n = 65$ representing the number of field data collection plots across all scars); b) delta TCB and RdNBR (rescaled by 0.01 to fit on the same graphs as TCB) for the combination of pre-burn image from June 20, 2009 and post-burn image for September 20, 2010 ($n = 52$ representing the number of field data collection plots across all scars).

Since the completeness of consumption of the inter tussock material is one of the metrics used in the ocular assessment of burn severity it relates well to the biophysical basis for the TCB assessment. With the full onset of greenness by mid-June, the signal from the exposed surface is overwhelmed by the regrowing vegetation and TCB expresses a weakened relationship to the BSI. At this stage spectral bands and indices (particularly SWIR 1.6 and RdNBR) which are more sensitive to the state of photosynthetically active vegetation produce better, although not quite as strong as late May TCB, relationships to burn severity. Additionally, field assessments of burn severity were based on the recovery of vegetation and were completed during June – August 2011 rather than immediately following the fire occurrence, thus potentially creating closer relationships with spectral signatures of the first post-fire season. It is important to note that the NIR is the second best spectral signal to use for burn severity measurement. However, when NIR is coupled with other bands to produce NVDI, NDSWIR, or NBR, the relationship between the resultant index and BSI weakens compared to that of the NIR alone (Appendix C).

The current operationally applied NBR served as the third strongest separator of burned/unburned signal. However, its ability to separate the gradient of burn severity was considerably lower than that of TCB. While NBR's amplitude between low and high burn severity (expressed through BSI) is the greatest, the scatter of individual values around the regression line is large (Fig. 7a). In comparison, although TCB offers a shallow slope of regression, the values are clustered tightly around the regression line and the change in the BSI/TCB relationship is considerably more linear compared to the BSI/NBR line. The same can be said about the RdNBR and delta TCB (Fig. 7b) assessment in relation to BSI measure. It is important to note that RdNBR provided the strongest relationship to BSI measurement when time periods of full greenness of vegetation (e.g. June and July) differencing was considered. However, the strength of the relationship was below r^2 of 0.8 and considerably below the closest relationship of r^2 0.87 given by delta TCB for differenced May scenes.

6. Conclusions

Our major results show that due to variability of spectral reflectance even across unburned regions of tundra, future operational methods should be developed from a large sample of known fire

events. We found that spectral characteristics of individual fire scars and surrounding areas may not be representative of broader conditions and that expected normalization of those conditions through conversion of spectral reflectance to spectral indices is insufficient. Similar to previous studies (e.g. Boelman et al., 2011) we confirmed that differencing pre-burn and post-burn imagery does not offer considerable improvements for mapping either the extent or severity of burning. Thus we suggest that single-date burned area mapping and burn severity estimates are feasible. Our results also showed that a single band threshold in NIR can be nearly as effective in burned area mapping within tussock tundra as complex multi-band indices, and more effective than currently applied NBR.

The analysis of temporal changes in fire-induced changes in surface reflectance shows a relatively short longevity of fire signatures in tundra (compared to Alaskan boreal forest) necessitating prompt post-fire mapping of burned areas and assessment of burn severity. The success in linking satellite observations to field measurement of burn severity is closely related to the timing of burning within the year and the date of the next available satellite image. Our results indicate that using TCB index during the early onset of greenness of the first post-fire season yields the strongest correlation between satellite-observed and field-measured assessments of burn severity.

Our results are limited to tussock tundra ecosystems within the Noatak Nature Preserve. More work is needed to assess if these findings are applicable to other tundra ecosystems. Future work will focus on extending this analysis to other tundra regions impacted by fire occurrence with an aim to develop a robust and flexible approach to providing consistent and contiguous assessments of fire impacts on North American tundra with a high degree of confidence.

Acknowledgements

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Appendix A

Clear-sky comparison of unburned samples across the study area using ANOVA and Scheffe F value statistics ($n = 1000$ pixels per scar; F critical at $p < 0.01$ is 4.109): scar1 – Rok North burn, scar2 – Kalukavik River burn, scar3 – Lake Tutusirok burn, scar4 – Nanielik Creek burn, scar5 – Makpik Creek burn, scar6 – Sidik Lake burn.

	Blue	Green	Red	NIR	SWIR 1.6	SWIR 2.2	NDVI	NDWI	NBR	TCB	TCG	TCW
23-Jun-10												
F (ANOVA)	325.3	209.4	187.9	156.5	169.4	205.5	97.8	137.7	155.8	237.4	95.0	185.1
F scars1&2	94.4	84.7	116.8	26.2	69.1	53.5	39.9	4.7	4.7	103.6	0.4	35.7
F scars1&3	183.2	102.6	48.8	14.5	2.1	0.1	12.1	6.4	8.2	36.7	0.5	4.1
F scars1&4	185.6	137.8	58.8	143.4	25.7	2.3	0.0	46.5	44.0	159.4	64.1	0.3
F scars1&5	263.5	161.9	102.0	52.4	65.2	31.9	22.6	0.0	0.2	138.4	5.8	18.3
F scars1&6	108.8	56.1	4.7	35.6	1.4	28.3	2.4	41.6	61.1	20.7	21.2	34.6
F scars2&3	14.6	0.9	14.6	1.7	47.0	57.1	8.0	22.1	25.4	17.0	0.0	64.0
F scars2&4	15.3	6.4	9.9	47.1	10.5	33.6	39.8	80.8	77.5	6.0	54.3	43.0
F scars2&5	42.5	12.4	0.5	4.5	0.1	2.8	2.4	4.2	6.6	2.5	3.1	2.9
F scars2&6	0.5	2.9	74.8	0.7	90.0	159.6	62.0	74.3	99.7	31.7	15.7	140.5
F scars3&4	0.0	2.6	0.5	66.8	13.0	3.1	12.1	18.4	14.2	43.2	53.7	2.1
F scars3&5	7.3	6.7	9.7	11.8	43.8	34.6	1.6	7.0	6.1	32.6	3.0	39.8
F scars3&6	9.6	7.0	23.2	4.7	6.9	25.8	25.4	15.4	24.5	2.3	15.4	14.8
F scars4&5	6.8	1.0	5.9	22.5	9.1	17.0	22.5	48.2	38.9	0.7	31.4	23.7
F scars4&6	10.2	18.0	30.3	36.2	38.9	46.7	2.4	0.1	1.4	65.3	11.6	28.0
F scars5&6	33.6	27.4	63.1	1.6	85.5	120.1	39.8	43.2	55.0	52.1	4.8	103.3

Appendix A (continued)

	Blue	Green	Red	NIR	SWIR 1.6	SWIR 2.2	NDVI	NDWI	NBR	TCB	TCG	TCW
20-Sep-10												
F (ANOVA)	323.9	245.5	336.3	155.9	122.2	75.8	45.4	69.9	84.5	203.0	71.2	73.9
F scars1&2	214.1	150.2	234.3	79.0	28.6	1.9	23.7	26.1	64.5	96.8	25.4	1.1
F scars1&3	169.0	17.2	30.9	13.7	0.0	0.7	0.8	0.1	19.4	10.5	5.2	4.1
F scars1&4	153.9	18.3	49.4	39.6	1.7	2.9	0.8	31.2	51.2	25.1	29.1	3.2
F scars1&5	179.1	20.6	59.5	41.5	7.3	0.0	0.7	24.7	40.9	34.6	25.4	0.0
F scars1&6	38.6	4.8	0.2	1.4	26.2	41.6	0.8	11.5	15.5	8.4	0.2	43.1
F scars2&3	2.7	65.8	95.0	26.9	30.7	4.8	16.0	29.2	13.1	43.5	7.6	9.4
F scars2&4	5.0	63.7	68.6	6.7	16.4	9.5	33.2	0.2	0.8	23.3	0.1	8.0
F scars2&5	1.6	59.5	57.7	6.0	7.1	1.5	32.5	0.0	2.7	15.6	0.0	1.0
F scars2&6	70.9	208.8	248.0	101.2	109.6	61.2	16.0	2.9	16.8	162.4	30.5	57.7
F scars3&4	0.4	0.0	2.1	6.7	2.2	0.8	3.1	34.6	7.5	3.1	9.7	0.1
F scars3&5	0.1	0.2	4.6	7.5	8.3	0.9	2.9	27.8	3.9	7.0	7.6	4.2
F scars3&6	46.0	40.2	36.0	23.8	24.3	31.8	0.0	13.6	0.2	37.7	7.7	20.5
F scars4&5	1.0	0.1	0.5	0.0	1.9	3.5	0.0	0.4	0.6	0.8	0.1	3.2
F scars4&6	38.3	41.8	55.7	55.7	41.2	22.4	3.1	4.8	10.3	62.7	34.5	22.8
F scars5&6	51.4	45.4	66.5	58.0	61.0	43.6	2.9	2.5	6.0	77.2	30.5	43.3
19-Jun-11												
F (ANOVA)	629.8	590.8	548.6	171.7	352.9	170.7	187.5	93.4	62.6	397.5	88.3	184.2
F scars1&2	329.3	274.2	266.2	56.7	128.8	21.0	56.1	0.6	11.4	171.9	3.6	24.9
F scars1&3	57.9	19.9	6.4	19.5	0.0	8.9	2.5	26.3	24.2	12.7	13.6	14.2
F scars1&4	38.9	14.0	3.5	61.6	0.1	15.9	22.9	68.1	56.1	28.7	58.7	22.2
F scars1&5	17.2	1.9	3.8	28.8	2.8	4.3	5.4	19.5	21.0	16.4	26.6	2.9
F scars1&6	29.1	47.3	39.2	5.7	51.0	60.1	10.1	6.6	7.3	36.2	0.1	54.2
F scars2&3	111.0	146.5	190.3	9.7	130.8	57.2	82.1	19.1	2.4	91.2	3.3	76.7
F scars2&4	141.9	164.4	208.9	0.1	121.3	73.4	150.5	56.2	16.9	60.1	33.4	94.1
F scars2&5	195.8	230.0	206.7	4.7	93.7	44.3	96.5	13.4	1.4	82.1	10.7	44.9
F scars2&6	554.3	549.3	509.8	98.3	341.8	152.2	113.8	3.3	0.5	365.8	2.4	152.5
F scars3&4	1.9	0.5	0.4	11.8	0.2	1.0	10.3	9.8	6.6	3.2	15.8	0.9
F scars3&5	12.0	9.4	0.3	0.9	3.1	0.8	0.6	0.5	0.1	0.2	2.1	4.2
F scars3&6	169.2	128.5	77.1	46.3	49.7	22.8	2.6	6.5	4.9	91.7	11.3	12.9
F scars4&5	4.3	5.5	0.0	6.1	1.8	3.7	6.0	14.7	8.5	1.7	6.3	9.0
F scars4&6	135.3	112.7	66.0	104.7	55.9	14.2	2.6	32.2	23.0	129.3	53.8	7.0
F scars5&6	91.2	68.4	67.3	60.1	77.6	32.2	0.7	3.4	3.5	101.3	23.3	31.9

Appendix B

Student t-test analysis of burned (n = 1000 random pixels per scar) and unburned (n = 1000 random pixels per scar) samples (black denotes pre-burn and red denotes post-burn conditions).

	4-Jun-09	20-Jun-09	16-Aug-09	30-May-10	23-Jun-10	9-Jul-10	20-Sep-10	25-May-11	19-Jun-11	22-Sep-11
Rok North burn										
blue	6.70	5.47	1.43	6.53	8.00	−10.94	−54.29		−35.29	−27.21
green	3.85	0.75	0.74	8.46	7.70	−59.91	−59.70		−55.52	−32.64
red	2.19	2.23	0.55	7.52	6.76	−36.93	−69.36		−32.68	−29.54
NIR	−4.30	−4.03	8.45	1.45	3.10	−175.61	−82.76		−96.36	−38.84
SWIR 1.6	−0.08	9.08	9.33	−1.61	12.63	−82.83	−75.20		−39.14	−45.95
SWIR 2.2	2.87	9.59	3.30	1.73	11.68	21.59	−24.76		39.74	−18.76
NDVI	−8.40	−4.34	4.52	−6.69	−4.87	−135.36	−34.21		−61.64	−16.83
NDWI	−8.53	−10.18	−3.10	3.28	−7.03	−132.72	−19.96		−70.68	1.49
NBR	−10.05	−9.35	0.83	−0.32	−7.71	−175.71	−33.99		−82.34	−12.29
TCB	−0.73	2.20	6.23	3.52	11.92	−121.58	−88.48		−71.41	−45.41
TCG	−7.10	−5.12	9.36	−2.64	−1.35	−194.85	−68.02		−101.66	−31.84
TCW	−1.43	−11.86	−7.80	3.08	−11.81	10.42	39.42		−20.95	29.05

(continued on next page)

Appendix B (continued)

	4-Jun-09	20-Jun-09	16-Aug-09	30-May-10	23-Jun-10	9-Jul-10	20-Sep-10	25-May-11	19-Jun-11	22-Sep-11
Kaluktavik River burn										
blue	−6.09	7.18	−3.67	2.55	5.00	2.25	−39.21	−35.06	−38.95	−10.17
green	−2.04	7.33	1.03	6.56	3.73	3.29	−50.68	−41.50	−65.38	−14.43
red	−1.46	7.67	1.73	5.71	5.35	2.78	−63.97	−51.31	−49.47	−17.49
NIR	5.53	−3.95	−6.10	4.09	−4.55	−1.92	−78.73	−74.22	−79.66	−39.07
SWIR 1.6	6.59	9.66	2.02	11.07	8.45	4.21	−52.22	−48.32	−58.11	−20.41
SWIR 2.2	2.05	8.24	0.37	7.65	7.72	4.49	−4.81	−8.34	48.65	0.92
NDVI	7.82	−8.51	−5.53	−2.31	−6.20	−4.01	−55.45	−58.67	−57.98	−21.54
NDWI	−1.50	−11.17	−9.32	−3.27	−9.88	−5.81	−82.07	−76.80	−76.92	−18.48
NBR	4.24	−8.69	−4.34	−2.05	−8.12	−4.68	−108.12	−86.31	−86.37	−36.88
TCB	3.86	3.86	−3.65	8.08	2.33	1.78	−64.11	−63.03	−80.31	−27.89
TCG	7.17	−7.15	−6.42	1.30	−6.46	−4.05	−83.26	−73.18	−73.50	−37.94
TCW	−5.90	−10.43	−3.20	−9.25	−9.90	−4.59	23.68	16.88	−26.50	6.17
Lake Tutusirok burn										
blue	0.32	4.71	10.15	−7.44	8.67	8.86	−39.70		−24.74	
green	3.26	6.22	14.47	−4.78	14.16	−23.54	−40.69		−38.70	
red	5.86	8.02	14.85	−2.99	12.66	−7.46	−45.85		−24.92	
NIR	12.57	6.52	20.66	1.97	7.60	−180.26	−46.53		−53.57	
SWIR 1.6	8.01	5.99	23.42	−0.66	10.83	−95.10	−59.23		−38.70	
SWIR 2.2	3.17	2.93	14.14	−1.54	7.40	27.27	−36.33		9.81	
NDVI	6.55	−5.16	−0.64	6.76	−5.42	−119.49	−10.89		−31.72	
NDWI	4.62	−0.30	10.92	6.15	0.31	−118.70	−54.72		−31.31	
NBR	6.97	0.59	1.11	6.58	−0.19	−161.19	−15.84		−41.74	
TCB	9.19	7.63	21.40	−1.03	16.57	−113.60	−57.34		−49.47	
TCG	11.93	2.36	13.60	6.09	2.25	−186.87	−34.45		−51.01	
TCW	−4.93	−2.81	−19.79	0.67	−5.15	12.42	47.42		3.84	
Nanielik Creek burn										
blue	−6.69		−24.54		2.44	−6.88	−45.72		−36.32	
green	−2.27		−15.84		−0.09	−29.77	−49.13		−49.97	
red	−2.69		−10.28		5.48	−20.10	−52.40		−31.07	
NIR	1.16		3.09		−12.80	−167.82	−66.32		−83.07	
SWIR 1.6	−4.81		1.86		−2.91	−99.98	−61.34		−50.08	
SWIR 2.2	−5.26		−10.74		7.27	19.13	−28.59		9.33	
NDVI	2.04		14.71		−14.45	−127.06	−25.39		−58.29	
NDWI	3.67		1.40		−12.66	−121.52	−9.21		−55.37	
NBR	3.17		15.44		−12.74	−165.09	−22.94		−62.29	
TCB	−2.14		−3.33		−6.80	−99.71	−71.16		−67.58	
TCG	2.85		10.37		−15.25	−199.10	−51.62		−86.03	
TCW	5.80		1.26		−4.21	27.67	39.33		6.20	

Appendix B (continued)

	4-Jun-09	20-Jun-09	16-Aug-09	30-May-10	23-Jun-10	9-Jul-10	20-Sep-10	25-May-11	19-Jun-11	22-Sep-11
Makpik Creek burn										
blue	5.13		7.34		11.77	−23.70	−44.95		−16.41	
green	10.87		13.28		14.52	−82.90	−43.65		−30.01	
red	14.02		13.75		18.79	−47.35	−42.10		−13.33	
NIR	17.58		15.33		9.42	−191.44	−53.07		−72.91	
SWIR 1.6	10.97		5.46		9.65	−107.79	−60.87		−22.33	
SWIR 2.2	4.92		4.84		8.75	15.00	−28.82		25.68	
NDVI	5.89		−1.18		−11.99	−160.78	−29.39		−63.41	
NDWI	12.45		11.14		0.40	−148.13	−10.86		−67.28	
NBR	12.71		6.96		−0.72	−190.02	−23.72		−73.45	
TCB	17.09		14.78		15.64	−147.58	−60.77		−43.59	
TCG	14.79		11.87		2.24	−198.64	−44.07		−83.93	
TCW	−2.97		0.92		−4.61	28.21	40.87		−15.80	
Sidik Lake burn										
blue	−3.45		−2.82		−90.52	11.00	−47.02	−20.03	−3.86	
green	1.93		0.34		−118.78	−16.11	−48.90	−27.56	−14.27	
red	1.71		−3.99		−58.91	−8.84	−43.17	−31.47	−12.65	
NIR	10.69		4.24		−173.17	−138.37	−48.84	−64.55	−57.13	
SWIR 1.6	5.18		−2.72		−79.34	−61.77	−50.27	−52.46	−49.66	
SWIR 2.2	0.81		−6.75		36.73	41.85	−28.22	−38.03	−7.92	
NDVI	5.90		7.85		−121.46	−94.02	−9.91	−36.59	−25.57	
NDWI	4.75		7.93		−137.96	−110.44	4.44	9.68	−11.92	
NBR	7.48		10.68		−151.42	−127.26	−9.49	−3.15	−33.95	
TCB	7.26		−0.23		−144.31	−92.39	−59.75	−63.35	−45.81	
TCG	9.29		6.89		−167.14	−133.90	−30.12	−50.86	−53.27	
TCW	−1.20		6.12		−15.13	−15.77	35.16	38.19	26.14	

Appendix C

Results of single-date regression analysis comparing spectral signatures and BSI measurements of burn severity ($p < 0.001$ is denoted with §, $p < 0.01$ is denoted with #, and $p < 0.05$ is denoted with *; n, representing the number of field plots, is shown individually for each available sample).

Date	n	Blue	Green	Red	NIR	SWIR 1.6	SWIR 2.2	NDVI	NDWI	NBR	TCB	TCG	TCW
<i>Single-date regressions</i>													
4-Jun-09	42	0.00	0.01	0.06	0.01	0.00	0.00	0.02	0.00	0.00	0.01	0.00	0.00
20-Jun-09	52	0.10*	0.06	0.14#	0.00	0.07	0.12*	0.18#	0.04	0.10*	0.06	0.04	0.09*
16-Aug-09	64	0.05	0.05	0.10*	0.02	0.05	0.08*	0.18§	0.01	0.10*	0.05	0.01	0.06
30-May-10	51	0.02	0.00	0.01	0.09*	0.19#	0.09*	0.14#	0.06	0.00	0.11*	0.10*	0.16#
23-Jun-10	59	0.01	0.07*	0.23§	0.00	0.01	0.02	0.24§	0.01	0.02	0.03	0.06	0.00
9-Jul-10	10	0.04	0.09	0.12	0.69#	0.59#	0.14	0.72§	0.57#	0.67#	0.61#	0.72§	0.17
20-Sep-10	65	0.34§	0.66§	0.77§	0.74§	0.75§	0.27§	0.40§	0.50§	0.70§	0.78§	0.67§	0.54§
25-May-11	50	0.72§	0.79§	0.81§	0.82§	0.73§	0.28§	0.63§	0.51§	0.69§	0.84§	0.75§	0.35§
19-Jun-11	65	0.68§	0.76§	0.78§	0.79§	0.80§	0.15#	0.30§	0.63§	0.79§	0.81§	0.75§	0.19§
22-Sep-11	33	0.09	0.16*	0.11	0.00	0.22#	0.45§	0.12*	0.13*	0.46§	0.16*	0.14*	0.39§

Appendix D

Results of delta (pre-burn – post-burn) regression analysis comparing spectral signatures and BSI measurements of burn severity ($p < 0.001$ is denoted with §, $p < 0.01$ is denoted with #, and $p < 0.05$ is denoted with *; n, representing the number of field plots, is shown individually for each available sample).

Dates	n	Blue	Green	Red	NIR	SWIR 1.5	SWIR 2.2	NDVI	NDWI	NBR	RdNBR	TCB	TCG	TCW
4-Jun-09 - 9-Jul-10	8	0.24	0.00	0.00	0.71#	0.53*	0.20	0.81§	0.46*	0.69#	0.75**	0.57*	0.77#	0.04
4-Jun-09 - 20-Sep-10	42	0.32§	0.45§	0.57§	0.47§	0.44§	0.03	0.36§	0.66§	0.73§	0.76§	0.48§	0.48§	0.23#
4-Jun-09 - 25-May-11	29	0.44§	0.52§	0.55§	0.62§	0.53§	0.14*	0.5§	0.61§	0.69§	0.70§	0.65§	0.42§	0.22#
4-Jun-09 - 16-Jun-11	42	0.26§	0.42§	0.38§	0.57§	0.27§	0.07	0.37§	0.66§	0.72§	0.77§	0.46§	0.53§	0
4-Jun-09 - 22-Sep-11	26	0.07	0.15	0.15*	0.01	0.15*	0.37§	0.16*	0.48§	0.54§	0.59§	0.13	0.09	0.28#
20-Jun-09 - 9-Jul-10	10	0.02	0.12	0.22	0.66#	0.61#	0.01	0.58#	0.49*	0.60#	0.65#	0.64#	0.66#	0.21
20-Jun-09 - 20-Sep-10	52	0.08*	0.45§	0.55§	0.76§	0.72§	0.03	0.43§	0.53§	0.67§	0.68§	0.72§	0.66§	0.4§
20-Jun-09 - 25-May-11	40	0.58§	0.81§	0.79§	0.83§	0.76§	0.28§	0.61§	0.43§	0.61§	0.62§	0.86§	0.68§	0.46§
20-Jun-09 - 16-Jun-11	52	0.29§	0.6§	0.63§	0.76§	0.77§	0.26§	0.46§	0.64§	0.77§	0.80§	0.76§	0.72§	0.04
20-Jun-09 - 22-Sep-11	33	0.11	0.18*	0.15*	0.04	0.25#	0.45§	0.13*	0.12*	0.41§	0.43§	0.24#	0.11	0.35§
16-Aug-09 - 9-Jul-10	10	0.03	0.22	0.27	0.76§	0.86§	0.01	0.35	0.48*	0.48*	0.58#	0.85§	0.49*	0.39*
16-Aug-09 - 20-Sep-10	64	0	0.06	0.18§	0.41§	0.4§	0	0.48§	0.58§	0.69§	0.68§	0.26§	0.54§	0.14#
16-Aug-09 - 25-May-11	50	0.49§	0.61§	0.63§	0.68§	0.59§	0.07	0.67§	0.63§	0.71§	0.71§	0.63§	0.63§	0.23§
16-Aug-09 - 16-Jun-11	64	0.03	0.12#	0.16#	0.48§	0.37§	0.15#	0.43§	0.64§	0.76§	0.77§	0.29§	0.58§	0
16-Aug-09 - 22-Sep-11	32	0.12*	0.13*	0.19*	0.06	0.23#	0.4§	0.22#	0.26#	0.49§	0.49§	0.18*	0.1	0.4§
30-May-10 - 9-Jul-10	8	0.16	0.01	0.06	0.68#	0.49*	0.27	0.69#	0.73#	0.85§	0.84§	0.53*	0.74#	0.11
30-May-10 - 20-Sep-10	51	0.16#	0.64§	0.75§	0.76§	0.74§	0.31§	0.4§	0.49§	0.68§	0.73§	0.78§	0.63§	0.57§
30-May-10 - 25-May-11	39	0.53§	0.82§	0.81§	0.83§	0.86§	0.6§	0.63§	0.53§	0.68§	0.72§	0.87§	0.72§	0.76§
30-May-10 - 16-Jun-11	51	0.37§	0.72§	0.72§	0.81§	0.71§	0	0.39§	0.51§	0.75§	0.81§	0.81§	0.73§	0.29§
30-May-10 - 22-Sep-11	28	0.07	0.14*	0.1	0	0.07	0.31#	0.22*	0.17*	0.44§	0.48§	0.09	0.32#	0.17*
23-Jun-10 - 9-Jul-10	10	0.07	0.07	0.06	0.63#	0.61#	0.08	0.56#	0.49*	0.64#	0.65#	0.58#	0.62#	0.24
23-Jun-10 - 20-Sep-10	59	0.3§	0.55§	0.66§	0.75§	0.77§	0.17#	0.46§	0.53§	0.67§	0.69§	0.77§	0.66§	0.56§
23-Jun-10 - 25-May-11	46	0.64§	0.76§	0.75§	0.83§	0.75§	0.24§	0.62§	0.54§	0.66§	0.67§	0.84§	0.7§	0.4§
23-Jun-10 - 16-Jun-11	59	0.71§	0.73§	0.79§	0.77§	0.85§	0.14#	0.44§	0.62§	0.75§	0.77§	0.83§	0.71§	0.27§
23-Jun-10 - 22-Sep-11	33	0.09	0.12*	0.16*	0.04	0.2#	0.38§	0.12*	0.09	0.34§	0.38§	0.21#	0.08	0.28#

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